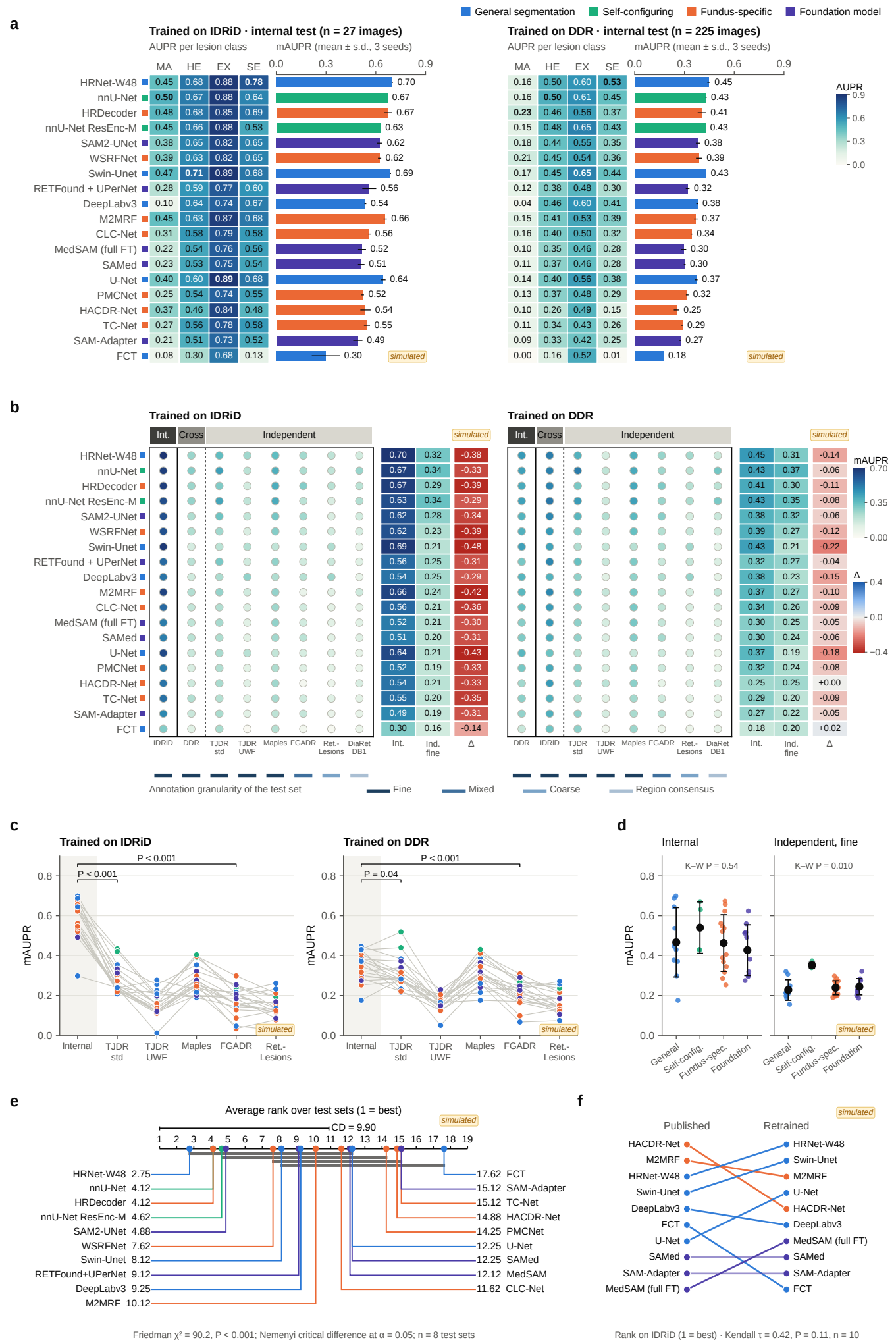

General segmentation
Self-configuring
Fundus-specific
Foundation model
a
Trained on IDRiD · internal test (n = 27 images)
AUPR per lesion class
mAUPR (mean ± s.d., 3 seeds)
MA HE EX SE
HRNet-W48 0.45 0.68 0.88 0.78 0.70
nnU-Net 0.50 0.67 0.88 0.64 0.67
HRDecoder 0.48 0.68 0.85 0.69 0.67
nnU-Net ResEnc-M 0.45 0.66 0.88 0.53 0.63
SAM2-UNet 0.38 0.65 0.82 0.65 0.62
WSRFNet 0.39 0.63 0.82 0.65 0.62
Swin-Unet 0.47 0.71 0.89 0.68 0.69
RETFound + UPerNet 0.28 0.59 0.77 0.60 0.56
DeepLabv3 0.10 0.64 0.74 0.67 0.54
M2MRF 0.45 0.63 0.87 0.68 0.66
CLC-Net 0.31 0.58 0.79 0.58 0.56
MedSAM (full FT) 0.22 0.54 0.76 0.56 0.52
SAMed 0.23 0.53 0.75 0.54 0.51
U-Net 0.40 0.60 0.89 0.68 0.64
PMCNet 0.25 0.54 0.74 0.55 0.52
HACDR-Net 0.37 0.46 0.84 0.48 0.54
TC-Net 0.27 0.56 0.78 0.58 0.55
SAM-Adapter 0.21 0.51 0.73 0.52 0.49
FCT 0.08 0.30 0.68 0.13 0.30
simulated
Trained on DDR · internal test (n = 225 images)
AUPR per lesion class
mAUPR (mean ± s.d., 3 seeds)
MA HE EX SE
0.16 0.50 0.60 0.53 0.45
0.16 0.50 0.61 0.45 0.43
0.23 0.46 0.56 0.37 0.41
0.15 0.48 0.65 0.43 0.43
0.18 0.44 0.55 0.35 0.38
0.21 0.45 0.54 0.36 0.39
0.17 0.45 0.65 0.44 0.43
0.12 0.38 0.48 0.30 0.32
0.04 0.46 0.60 0.41 0.38
0.15 0.41 0.53 0.39 0.37
0.16 0.40 0.50 0.32 0.34
0.10 0.35 0.46 0.28 0.30
0.11 0.37 0.46 0.28 0.30
0.14 0.40 0.56 0.38 0.37
0.13 0.37 0.48 0.29 0.32
0.10 0.26 0.49 0.15 0.25
0.11 0.34 0.43 0.26 0.29
0.09 0.33 0.42 0.25 0.27
0.00 0.16 0.52 0.01 0.18
simulated
AUPR
b
Trained on IDRiD
Int. Cross Independent
IDRiD DDR TJDR std TJDR UWF Maples FGADR Ret.-Lesions DiaRet DB1
Int. Ind. fine Δ
HRNet-W48 0.70 0.32 -0.38
nnU-Net 0.67 0.34 -0.33
HRDecoder 0.67 0.29 -0.39
nnU-Net ResEnc-M 0.63 0.34 -0.29
SAM2-UNet 0.62 0.28 -0.34
WSRFNet 0.62 0.23 -0.39
Swin-Unet 0.69 0.21 -0.48
RETFound + UPerNet 0.56 0.25 -0.31
DeepLabv3 0.54 0.25 -0.29
M2MRF 0.66 0.24 -0.42
CLC-Net 0.56 0.21 -0.36
MedSAM (full FT) 0.52 0.21 -0.30
SAMed 0.51 0.20 -0.31
U-Net 0.64 0.21 -0.43
PMCNet 0.52 0.19 -0.33
HACDR-Net 0.54 0.21 -0.33
TC-Net 0.55 0.20 -0.35
SAM-Adapter 0.49 0.19 -0.31
FCT 0.30 0.16 -0.14
simulated
Trained on DDR
Int. Cross Independent
DDR IDRiD TJDR std TJDR UWF Maples FGADR Ret.-Lesions DiaRet DB1
Int. Ind. fine Δ
0.45 0.31 -0.14
0.43 0.37 -0.06
0.41 0.30 -0.11
0.43 0.35 -0.08
0.38 0.32 -0.06
0.39 0.27 -0.12
0.43 0.21 -0.22
0.32 0.27 -0.04
0.38 0.23 -0.15
0.37 0.27 -0.10
0.34 0.26 -0.09
0.30 0.25 -0.05
0.30 0.24 -0.06
0.37 0.19 -0.18
0.32 0.24 -0.08
0.25 0.25 +0.00
0.29 0.20 -0.09
0.27 0.22 -0.05
0.18 0.20 +0.02
simulated
mAUPR
Δ
Annotation granularity of the test set
Fine
Mixed
Coarse
Region consensus
c
Trained on IDRiD
P < 0.001
P < 0.001
mAUPR
Internal TJDR std TJDR UWF Maples FGADR Ret.-Lesions
simulated
Trained on DDR
P < 0.001
P = 0.04
mAUPR
Internal TJDR std TJDR UWF Maples FGADR Ret.-Lesions
simulated
d
Internal
K–W P = 0.54
Independent, fine
K–W P = 0.010
mAUPR
General Self-config. Fundus-spec. Foundation
simulated
e
Average rank over test sets (1 = best)
CD = 9.90
simulated
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
HRNet-W48 2.75
nnU-Net 4.12
HRDecoder 4.12
nnU-Net ResEnc-M 4.62
SAM2-UNet 4.88
WSRFNet 7.62
Swin-Unet 8.12
RETFound+UPerNet 9.12
DeepLabv3 9.25
M2MRF 10.12
17.62 FCT
15.12 SAM-Adapter
15.12 TC-Net
14.88 HACDR-Net
14.25 PMCNet
12.25 U-Net
12.25 SAMed
12.12 MedSAM
11.62 CLC-Net
Friedman χ² = 90.2, P < 0.001; Nemenyi critical difference at α = 0.05; n = 8 test sets
f
simulated
Published
Retrained
HACDR-Net
M2MRF
HRNet-W48
Swin-Unet
DeepLabv3
FCT
U-Net
SAMed
SAM-Adapter
MedSAM (full FT)
HRNet-W48
Swin-Unet
M2MRF
U-Net
HACDR-Net
DeepLabv3
MedSAM (full FT)
SAMed
SAM-Adapter
FCT
Rank on IDRiD (1 = best) · Kendall τ = 0.42, P = 0.11, n = 10